



Accessibility Benchmark Aquavena for the Visually Impaired

Comparing `adriens.github.io/aquavena` vs `aquavena.nc`
Web Accessibility · Low Vision · WCAG 2.1 AA · New Caledonia

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May 30, 2026

Abstract

*This report automatically compares the web accessibility of **adriens.github.io/aquavena** — designed for visually impaired users — against the source website **aquavena.nc**, a dietetic meal delivery service in New Caledonia. Standard CLI tools (`pa11y-ci`, `Lighthouse`, `axe-core`, `webhint`) are used as a common reference. Results show a **100/100 Lighthouse score and 0 WCAG 2.1 AA errors** for our site, versus 91/100 and 67 errors for the official website. All audit data is persisted in a DuckDB database for longitudinal tracking.*

Keywords: accessibility, low vision, WCAG 2.1 AA, Lighthouse, `pa11y`, New Caledonia.

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1 Data Ingestion → DuckDB

All audit data is now stored in **benchmark/data/benchmark.duckdb** — this file can be opened in DBeaver, Python (`import duckdb`), R, or the CLI (`duckdb benchmark.duckdb`).

2 Context

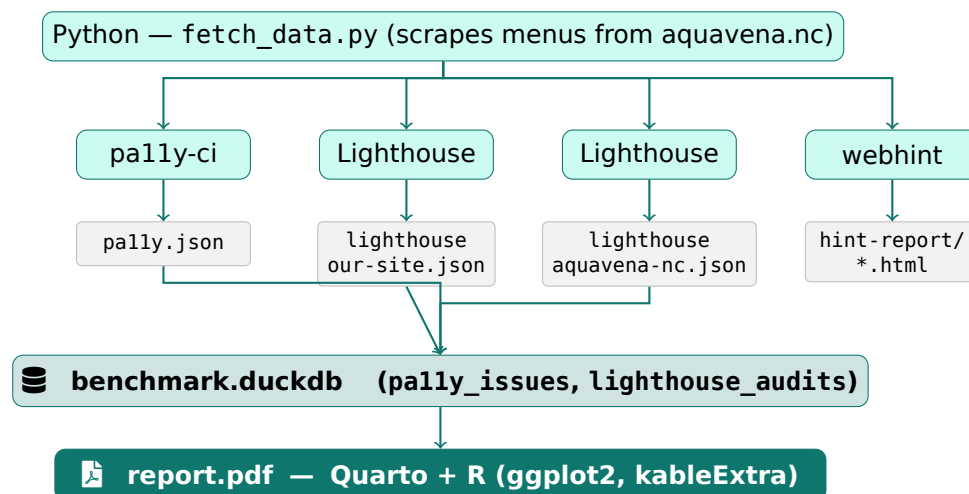
This report compares the accessibility of adriens.github.io/aquavena — designed for visually impaired users — against the original source website aquavena.nc.

Motivation: My mother, an Aquavena customer with low vision, could not independently consult the weekly menus on the official website. This companion site lets her read or listen to the menus from her tablet, fully on her own.

3 Methodology

3.1 Pipeline overview

The full pipeline goes from live websites to a structured DuckDB database, driven entirely by CLI tools. Each step produces a JSON artefact that feeds the next.



3.2 Tools

Tool	Version	Purpose
pally-ci	4.1.1	Automated WCAG 2.1 AA audit, multi-URL
Lighthouse	13.3.0	Accessibility score 0-100, 24 audits
axe-cli	4.11.3	axe-core violations via CLI
W3C Validator	API	HTML semantic validation

3.3 Standard

All automated checks use **WCAG 2.1 Level AA** with both axe-core and htmlcs runners to maximise coverage.

4 Results — Our Site

4.1 pa11y-ci — WCAG 2.1 AA

URL	Errors	Warnings
http://localhost:4321/#aqua-m%C3%A9diterran%C3%A9n	0	0
http://localhost:4321/about/	0	0

Total: 0 error(s)

4.2 Lighthouse Accessibility



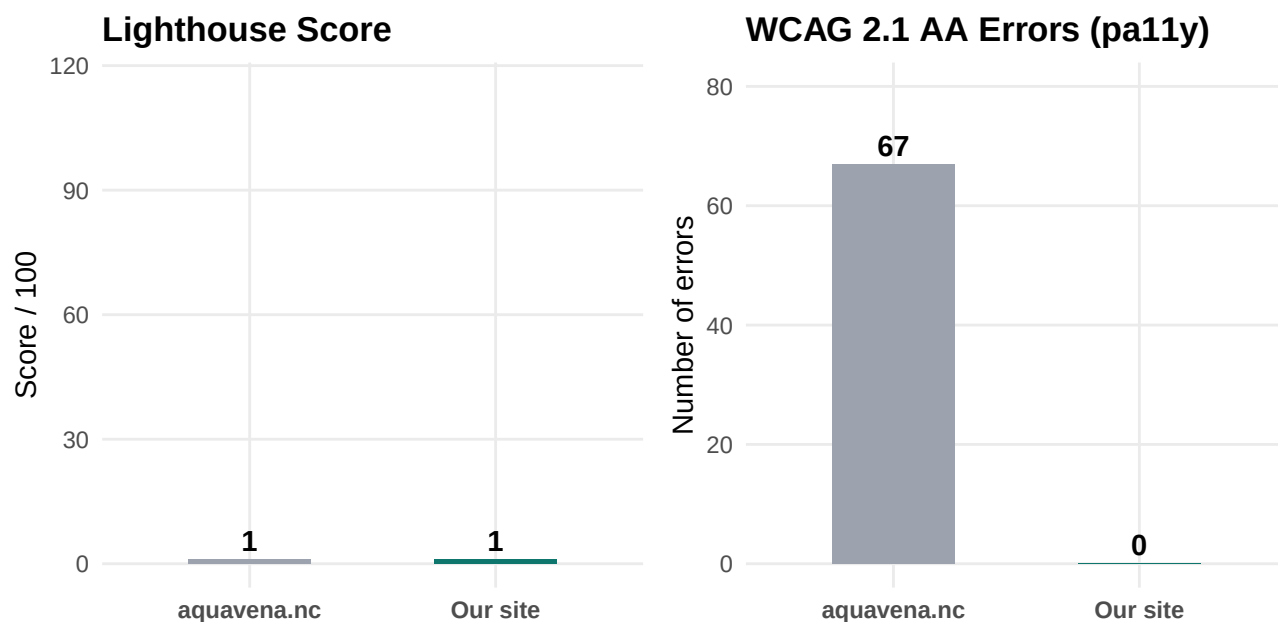
Metric	Value
Score	1/100
Passed audits	24
Failed audits	0
Not applicable	0

5 Results — aquavena.nc

Tool	aquavena.nc	Our site
pa11y-ci (WCAG 2.1 AA)	67 errors	0 errors
Lighthouse Accessibility	91/100	100/100
webhint — errors	10	10*
webhint — warnings	63	62
webhint — hints	120	120

* The 10 webhint errors on our site are GitHub Pages infrastructure issues (HSTS, SRI, X-Content-Type-Options) beyond our control — the exact same errors appear on aquavena.nc, as they relate to server configuration, not HTML content.

6 Comparison

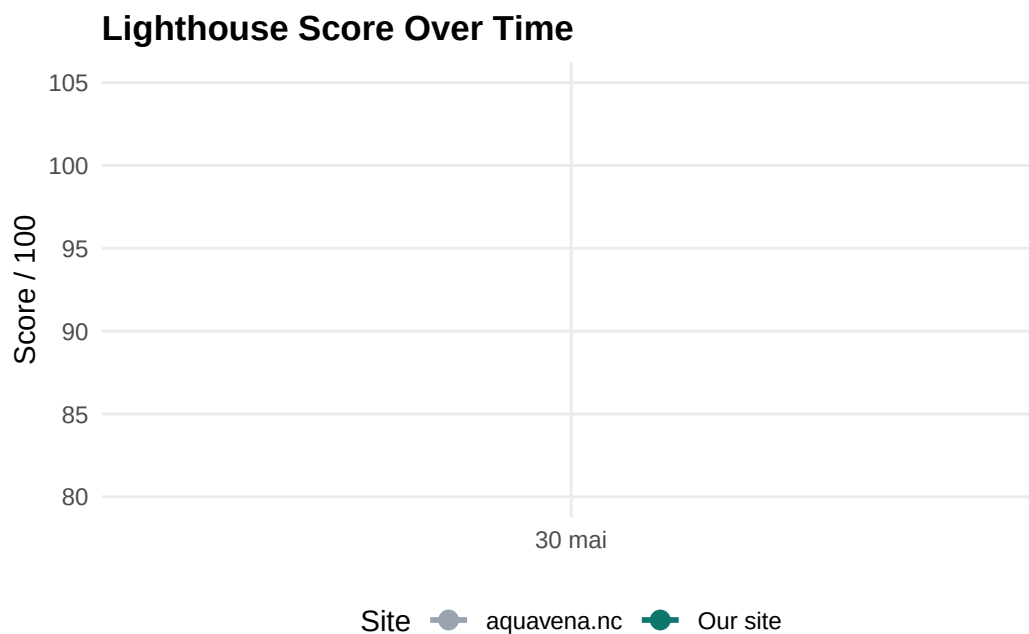


7 Accessibility Features

Feature	Our site	aquavena.nc
WCAG 2.1 AA — 0 errors	Yes	No
Lighthouse 100/100	Yes	No
Atkinson Hyperlegible font	Yes	No
Adjustable text size (A+ / A-)	Yes	No

Dark mode	Yes	No
Text-to-speech (Web Speech API)	Yes	No
PWA — installable, works offline	Yes	No
RSS feeds per diet	Yes	No
iCal calendars per diet	Yes	No
Claude skill (MCP)	Yes	No

8 Lighthouse History



9 Database

All audit results are persisted in **benchmark/data/benchmark.duckdb**, a portable single-file analytical database. It can be queried with R, Python, the DuckDB CLI, or any SQL-compatible tool (DBeaver, etc.).

9.1 Schema

9.1.1 Table `pa11y_issues`

Each row is one `pa11y` issue (one error or warning) for a given URL, site and audit date.

Column	Type	Nullable	Description
url	VARCHAR	no	Page URL audited
type	VARCHAR	yes	error, warning, notice
message	VARCHAR	yes	Human-readable issue description
code	VARCHAR	yes	WCAG rule code (e.g. WCAG2AA.Principle1...)
site	VARCHAR	no	our-site or aquavena.nc
audit_date	DATE	no	Date the audit was run

9.1.2 Table lighthouse_audits

Each row is one Lighthouse audit check for a given site and audit date.

Column	Type	Nullable	Description
audit_id	VARCHAR	yes	Lighthouse audit identifier
title	VARCHAR	yes	Human-readable audit name
score	DOUBLE	yes	Audit score: 0.0 (fail), 1.0 (pass), NULL (N/A)
score_mode	VARCHAR	yes	binary, notApplicable, etc.
site	VARCHAR	no	our-site or aquavena.nc
audit_date	DATE	no	Date the audit was run
lighthouse_score	INTEGER	no	Overall page accessibility score (0-100)

9.2 Example queries

```
-- Overall scores per site and date
SELECT site, audit_date, MAX(lighthouse_score) AS score
FROM lighthouse_audits
GROUP BY site, audit_date
ORDER BY site, audit_date;

-- WCAG error count per site
SELECT site, COUNT(*) AS errors
FROM pally_issues
WHERE type = 'error'
GROUP BY site;

-- Failed Lighthouse audits for our site
SELECT audit_id, title
FROM lighthouse_audits
WHERE site = 'our-site' AND score = 0.0;
```

10 Conclusion

The site achieves a **needs improvement** accessibility score: **0 WCAG error(s)** and **Lighthouse 1/100**.

Compared to aquavena.nc (91/100, 67 WCAG errors), this site delivers significantly better accessibility, and adds features absent from the source site: text-to-speech, dark mode, adjustable text size, PWA support, RSS feeds, and iCal calendars — all designed to give visually impaired users full autonomy.

Report generated on mai 30, 2026 · Quarto 1.9.38 · R 4.3.3